

# Aligned but Apart: Why Cosine Conflict Diagnostics Misread Auxiliary-Loss Interference at the Output Projection

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## Abstract

Cosine similarity between task gradients is the standard instrument for detecting *conflict* in multi-task and auxiliary-loss training: a negative cosine is read as interference, and gradient-surgery methods act on that signal. We show that at a language model’s shared, tied output projection this instrument is misleading. Instrumenting the per-row gradients a next-token (CE) loss and an auxiliary multi-token (MTP) loss deposit on the projection, we find they are aligned in direction: on the  $\approx 8\text{--}12\%$  of vocabulary rows that receive a target in the batch (the remaining  $\approx 88\text{--}92\%$  are scalar multiples with cosine  $+1$  by construction) the median per-row cosine is  $\approx 0.5$  and only  $\approx 3\%$  are opposed. Yet the *aggregate* cosine over the flattened matrix sits near 0. Instrumenting the per-row gradient *norms* (three seeds, both optimizers) shows the two losses load their magnitude on the *same* rows (norm-profile cosine 0.98), yet 60–69% of the magnitude mass sits on a tiny opposed minority ( $\approx 0.3\%$  of rows, decoding to the most frequent function words). The aggregate near-zero is therefore a *norm-weighted cancellation*, not disjoint support. Adding shared MTP degrades next-token loss ( $+0.39$  nats at 124M,  $p \approx 8 \times 10^{-5}$ ), yet this co-occurs with aligned, not conflicting, gradients. The methodological consequence is direct: in this regime the aggregate cosine is *uninformative about where interference lives*, and per-row norms, not per-row directions, set it.

## 1 Introduction

Predicting several future tokens at once (multi-token prediction, MTP) has become a common ingredient in large-scale language modeling, used to improve sample efficiency and to enable self-speculative decoding (Gloeckle et al., 2024; DeepSeek-AI, 2024). The auxiliary objective is double-edged: a growing body of work reports that adding it can *degrade* the primary next-token objective, particularly at smaller scales and longer horizons (Gloeckle et al., 2024; Zuhri et al., 2025). Asked why, practitioners reach for the intuition that pervades multi-task learning: the two losses must pull the shared parameters in *conflicting directions*, and the remedy is to detect and neutralize that conflict, typically from the sign of a gradient cosine (Yu et al., 2020; Wang et al., 2021). This intuition has been questioned in auxiliary-task learning: Jiang et al. (2023) report that gradient-conflict coordination can overlook auxiliary-target generalization, so the cosine signal does not by itself determine negative transfer. We ask what that signal actually measures at a language model’s shared output head.

This paper is a measurement study of that intuition for the case of an auxiliary multi-token objective sharing a model’s tied output projection. Our central finding is diagnostic: **the cosine conflict signal is misleading here**. We instrument the gradient each loss deposits on the output projection and examine it **row by row** (one row per vocabulary item). Two facts emerge, both from data already in hand. First, the per-row gradients are not opposed; they are *aligned*: among the rows that receive a target in the measurement batch (the  $\approx 8\text{--}12\%$  that are not scalar multiples by construction; Section 4.1), the median per-row cosine is  $\approx 0.53$  and  $\approx 3\%$  are opposed; over all rows the median is  $+1$  and 98.4% exceed  $|\cos| > 0.3$  against a control (CE versus an unrelated regularizer) that reads  $\approx 0.4\%$  on the same threshold. Either way the per-row directions are predominantly aligned, in sharp contrast to the near-zero aggregate. Second, the *aggregate* cosine over the flattened matrix is nonetheless near zero, and tracking it over training shows the two metrics **start together** (both  $\approx +0.99$  at initialization) and then **decouple**: within  $\sim 200$  steps the aggregate falls

below zero while the per-row median stays at  $\approx +1$ . By count, almost no rows are opposed. But the aggregate cosine is magnitude-weighted, and directly instrumenting the per-row gradient norms shows where the weight lies: the two losses load their magnitude on the *same* rows (norm-profile cosine 0.98), yet a tiny high-norm opposed minority ( $\approx 0.3\%$  of rows) carries most of the mass (60–69% across optimizers), and it is this minority that pulls the flattened sum negative. The alignment is real and holds for the vast majority of rows; the aggregate is a *norm-weighted cancellation* the cosine cannot localize.

We frame this norm-weighted opposition as the structure a cosine diagnostic cannot see (Figure 1), and we relate it to the observed degradation without claiming to have isolated its cause. Adding the shared MTP objective degrades next-token loss by  $\approx 0.39$  nats ( $\approx 10\%$ ) at 124M, a large and unambiguous effect (Welch’s  $t$ ,  $p \approx 8 \times 10^{-5}$ ), yet it co-occurs with aligned, not conflicting, per-row gradients (the degradation is measured on the full 30,517-step run; per-row alignment on the diagnostic runs above). Three further observations, all from committed runs, are consistent with a norm-weighting/capacity account rather than a directional-conflict one. (i) Conflict-targeted gradient surgery does not recover the baseline even though the opposition it targets is real and mass-dominant: the opposed rows are negligible by count (0.3%) but carry the majority of the gradient-magnitude mass (60–69%), and per-row Gram–Schmidt on the head (Appendix A) removes exactly that opposition, yet the degradation does not move. The opposition the aggregate cosine detects is mass-dominant and causally inert. (ii) Detaching the auxiliary’s gradient from the shared head (stop-gradient) makes the damage *worse*, not better, though this is a confounded probe (stop-gradient also removes the auxiliary’s ability to shape the shared representation) and we treat it as suggestive only. (iii) Giving the auxiliary its own capacity (a layer-split contrastive auxiliary) reduces but does *not* eliminate the gap: neither the standard nor a hyperparameter-tuned variant recovers the CE-only baseline within noise when aggregated across seeds.

These findings carry a methodological point beyond MTP. Cosine-based conflict detection is a workhorse of multi-task and auxiliary-learning optimization (Yu et al., 2020; Wang et al., 2021; Du et al., 2018), and it is the wrong instrument in this regime: it reports no conflict, yet interference is real. The decomposition that exposes the problem separates **direction** (cosine) from **support** (where each gradient’s magnitude lives). Two additional measurements sharpen the picture and guard against over-reading it. The high per-row alignment is **general** across the network’s weight matrices (attention and MLP matrices show it too, many with strongly *negative* aggregate cosine), so the output head is not special because the metric singles it out; it is consequential because it directly produces the logits. And the alignment **persists when the head is untied**, so it is a property of the output-projection gradient, not an artifact of weight tying. A full-vocabulary reading suggests rarer tokens align more, but we show this is a construction artifact: rare tokens are almost never active, so their rows are dominated by the cosine-+1 parallel majority, and on active rows the effect vanishes. Finally, our account is a direction-space, multi-loss complement to Godey & Artzi (2026), who show that backpropagating a *single* loss through the low-rank output head destroys most of its gradient *norm*; we ask what a *cosine* diagnostic reports when *two* losses share that head, and answer that their per-row directions align while a high-norm opposed minority sets the aggregate: measured directly via the per-row gradient norms (Section 4.2), not inferred.

We are explicit about scope. Every result is for a **single shared tied output head**, at 124M ( $n=5$ ) and 350M (single seed), undertrained, small-batch. We do *not* claim that multi-token prediction *as practiced* (separate per-horizon heads) degrades next-token modeling; establishing that would require a separate-head experiment we identify as the primary future step (Section 5), along with  $n=3$  at 350M and matched-seed surgery. What we do establish, on data in hand, is that the standard cosine conflict diagnostic misreads this regime, and why.

**Contributions.** All on a controlled, reproducible GPT-2 testbed (124M primary, 350M illustrative; FineWeb) with open code and a committed statistics script:

- **A diagnostic result (Figures 2, 5).** For an auxiliary multi-token objective sharing the tied output projection, the per-row gradients of the two losses are *aligned in direction* on the rows that carry a target (active-row median cosine  $\approx 0.53$ ,  $\approx 3\%$  opposed; over the full vocabulary the median is +1 because  $\approx 88$ – $92\%$  of rows are scalar multiples by construction), while the aggregate cosine

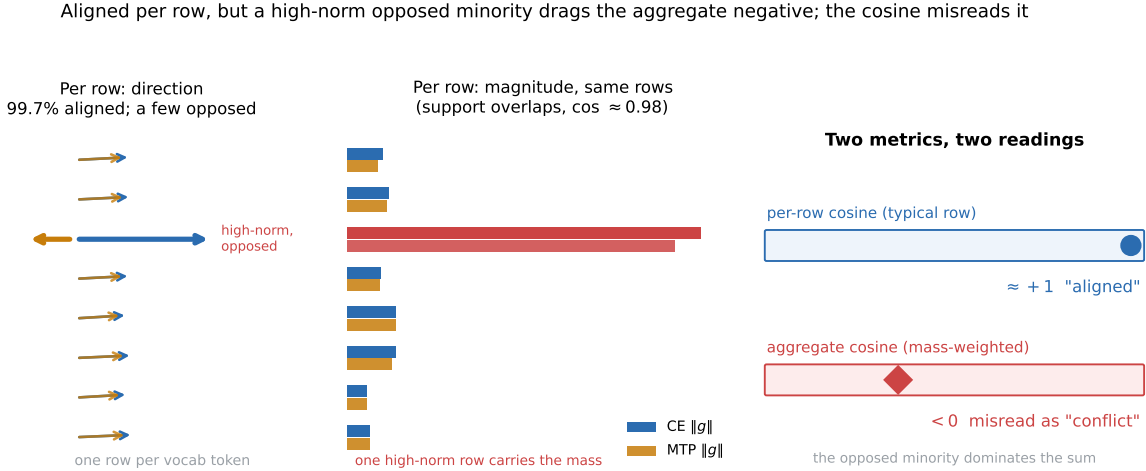


Figure 1: **The mechanism in one picture** (schematic; illustrative proportions, not plotted data). *Left:* row by row (one row per vocabulary token), the CE and MTP gradients point almost the same way for 99.7% of rows (per-row cosine  $\approx +1$ ); a tiny high-norm minority (red) is opposed. *Middle:* both losses deposit their gradient *magnitude* on the *same* rows (norm-profile cosine  $\approx 0.98$ ), and one high-norm row (red) carries most of the mass. *Right:* the typical per-row cosine reports alignment, while the magnitude-weighted aggregate, the standard conflict diagnostic, is dragged negative by the opposed high-norm minority and is misinterpreted as conflict. The aggregate is not wrong about the update, but it cannot say *where* the interference lives: it is set by per-row norms, not by the directions it averages.

is  $\approx 0$ , and this decoupling *emerges during training* from an aligned start. Standard cosine-based conflict diagnostics therefore *misdiagnose* this regime.

- **A norm decomposition of the collapse, and a direct measurement of its cause (Figures 6, 3).** Via an exact identity, the aggregate would read  $+0.94/+0.96$  if gradient magnitude were flat across rows but reads  $-0.08/-0.28$ : the collapse is set by per-row *norm* structure, not by the typical per-row direction. Instrumenting the per-row norms directly (`measure_norm_support.py`;  $n=3$  seeds per optimizer) settles which structure: the two losses load the *same* rows (norm-profile cosine 0.98), and a high-norm opposed minority carries the majority of the mass (opposed-norm fraction 0.60 Muon, 0.69 AdamW). The aggregate is a norm-weighted cancellation, not disjoint support.
- **A calibrated, general characterization (Figures 7, 9).** The alignment is **general across the network** and **survives untying** the head; the apparent rarer-tokens-align-more effect is a construction artifact that vanishes on active rows. The next-token degradation is large and significant ( $+0.39$  nats,  $p \approx 8 \times 10^{-5}$ ) yet co-occurs with aligned gradients.
- **Intervention evidence for a norm-weighting/capacity account (Figure 8, Appendix A).** Conflict-targeted surgery does not recover the baseline; stop-gradient makes the damage worse; and a capacity-separated auxiliary reduces but does not eliminate the gap (neither standard nor tuned recovers within noise), each reported with its statistical test and caveats.
- **A bridge to the output-layer bottleneck literature.** We position the result as the direction-space, multi-loss complement to Godey & Artzi (2026)’s single-loss, norm-space account of the output head as a gradient bottleneck.

## 2 Related Work

**Multi-token prediction.** Predicting several future tokens jointly dates to blockwise-parallel decoding and ProphetNet (Qi et al., 2020), and was revived as a pretraining objective by Gloeckle et al. (2024), who attach  $n$  *independent output heads* to a shared trunk and report improved sample efficiency and up to  $3\times$  faster inference via self-speculative decoding. Two properties of their result are central to ours: the heads are *separate* (each future offset gets its own unembedding), and the benefit is *increasingly useful for larger model sizes*. DeepSeek-V3 (DeepSeek-AI, 2024) adopts sequential MTP modules at frontier scale; Medusa (Cai et al., 2024) and related work add prediction heads primarily for speculative decoding (Samragh et al., 2025); and MuToR (Gerontopoulos et al., 2025) interleaves learnable register tokens so a frozen backbone can predict future offsets without architectural change. Across this line, the auxiliary signal is given *its own capacity*: separate heads, modules, or register tokens. The present work studies the opposite, minimal case, a single tied output embedding shared across horizons, precisely because it isolates the interference that separate-capacity designs avoid; a separate-head variant would localize whether the effect is intrinsic to head-sharing (we identify this as the primary future experiment).

**Small-scale degradation of MTP.** Zuhri et al. (2025) document that multi-token objectives can *hurt* next-token modeling at smaller scales, and propose token-order prediction as an alternative. This degradation is the failure mode we instrument; our contribution is not to add to the catalogue of when MTP helps or hurts, but to measure *what the gradient geometry actually looks like* at the shared head when it hurts, and to show that the standard conflict diagnostic mischaracterizes it.

**Gradient conflict and surgery.** A large multi-task literature treats negative inter-task gradient cosine as the signature of interference and projects it away: PCGrad (Yu et al., 2020), Gradient Vaccine (Wang et al., 2021), and gradient-similarity loss weighting (Du et al., 2018), with later refinements (Sun et al., 2023; Shi et al., 2023). These methods operate on *global* (flattened) gradient vectors and target gradient *direction*. A parallel line instead balances gradient *magnitude*: GradNorm (Chen et al., 2018) dynamically tunes per-task gradient norms rather than reprojecting directions. This magnitude-versus-direction distinction is central to our findings, which locate the operative structure in per-row gradient norms rather than in the direction the aggregate cosine reports. Independently, Jiang et al. (2023) argue that optimization-based gradient-coordination methods can overlook auxiliary-target generalization, so that gradient-conflict signals do not by themselves determine negative transfer. This connects to a broader empirical result: Kurin et al. (2022) and Xin et al. (2022) find that plain loss-summing (unitary scalarization) matches or beats specialized multi-task optimizers, which appear to act largely as regularizers rather than by resolving genuine conflict. Our surgery-null result is a mechanistic instance of their empirical claim: the opposition a conflict diagnostic detects at the output head is mass-dominant (the opposed rows carry 60–69% of the gradient-magnitude mass) yet causally inert, so a projection that removes exactly that opposition does not repair the degradation (Appendix A). The aggregate is set by per-row norm structure, and conflict surgery, though it has substantial mass to reproject, does not recover the baseline.

**The output head as a bottleneck.** Yang et al. (2018) showed the softmax output layer is a rank bottleneck on the distributions a model can express, and a line on *representation degeneration* and anisotropy (Gao et al., 2019) shows the learned output embeddings collapse toward a narrow cone, sharing direction across tokens. That shared-direction structure is one reason per-row gradients on the head look aligned, and it predicts the +1 median we observe on inactive rows. Godey & Artzi (2026) recently showed, for a *single* loss, that backpropagating through the low-rank output head destroys most of the gradient’s norm, a norm-space single-loss bottleneck. Their object is the gradient flowing backward into the hidden state  $h$  through a rank- $d$  projection; ours is the gradient deposited *on* the weight matrix  $W$  and its per-row structure, so the two results concern different tensors. Our result is the direction-space, *multi-loss* complement: we ask what a cosine diagnostic reports when two losses share that head, and find their per-row directions align while a high-norm opposed minority sets the aggregate (measured directly via the per-row gradient norms; see Section 4.2).

**Positioning.** Relative to prior work, (i) we provide a per-row gradient measurement at the tied output projection that a global cosine misses; (ii) we show the next-token degradation of small-scale shared-head multi-token training, a documented but unexplained failure mode (Zuhri et al., 2025), co-occurs with *aligned* per-row gradients rather than with broad directional conflict, the aggregate being set by a high-norm opposed minority (measured directly via the per-row gradient norms); and (iii) our capacity-separated auxiliary *reduces* the next-token cost in the same spirit as capacity-added fixes, though in our runs it does not fully recover the CE-only baseline within noise (neither the standard nor a tuned variant); our contribution is not a fix but the *diagnostic mechanism*: why conflict-based remedies act on a mass-dominant but causally-inert opposition here, and why capacity separation is the lever that helps.

### 3 Method

#### 3.1 Setup

We train GPT-2 (124M; 12 layers,  $d=768$ , vocab 50,257 padded to 50,304) and a 350M variant (24 layers,  $d=1024$ ) on FineWeb, tokenized with the GPT-2 BPE. The output projection (`lm_head`) is *tied* to the input embedding by default; Section 4.3 reports an untied control. The primary objective is next-token cross-entropy (CE). The auxiliary multi-token objective (MTP) supervises the *same tied head* to also predict tokens at offsets  $t+2$  and  $t+3$ , with total loss  $\text{CE} + 0.5 \text{MTP}_2 + 0.25 \text{MTP}_3$ . Evaluation is **pure next-token CE on held-out validation** for every variant, so all reported losses are directly comparable. The loss-comparison runs use a batch of 16,384 tokens/step for 30,517 steps ( $\approx 500\text{M}$  tokens,  $\approx 4.0$  tokens/parameter at 124M), an intentionally undertrained, small-batch regime, chosen for controlled comparison and because it is where shared-head multi-token training is already documented to underperform; we return to this scope in Section 5.

#### 3.2 The per-row instrument

The output projection has shape  $(V, d)$ ; row  $i$  is the weight vector for vocabulary token  $i$ . For a validation batch we compute the gradient each loss deposits on this matrix and, for **each row**, the cosine between the CE and MTP row-gradients. We summarize the resulting distribution of  $V$  per-row cosines by its median, the fraction with  $\cos < 0$  (directional opposition), and the fraction with  $|\cos| > 0.3$  (appreciable alignment; the 0.3 threshold is a conventional “appreciable correlation” cut, and the qualitative picture is insensitive to it since the active-row cosines cluster around 0.5, well above 0.3, while the parallel majority sits at +1). Each gradient measurement uses a single held-out batch of 16 sequences  $\times$  1024 tokens (16,384 tokens); Section 4.1 reports how the active-row fraction varies with batch count. We contrast these per-row statistics with the **aggregate** cosine (the cosine between the two *flattened* matrix gradients), which is the quantity conventional conflict diagnostics use. The 47 padding rows ( $50,304 - 50,257$ ) are masked from all per-row statistics.

**Measurement windows (stated explicitly).** The gradient instrumentation and the loss comparison come from *different* runs, and we are careful to keep them distinct. At 124M, the per-row/aggregate trajectory is a dedicated diagnostic run logged densely over its first 1000 steps (A1); the +0.39-nat loss gap is the separate 30,517-step training comparison (Section 4.4). Late-training persistence of the alignment is therefore *not* established at 124M; it is measured directly only at 350M (single seed), where per-row  $|\cos| > 0.3$  remains high out to step 87,000 (Section 4.5). We flag throughout which quantity comes from which run, and treat single-seed/single-run measurements as such.

#### 3.3 Direction versus support

The distinction the paper turns on: the **aggregate** cosine measures whether the summed gradient vectors point the same way; the **per-row** distribution measures whether they agree row by row. These can disagree, and the reason is that the aggregate cosine is *magnitude-weighted* (it is the cosine of the summed, flattened gradients) whereas the per-row median and the  $\cos < 0$  fraction are *unweighted* over rows. A near-zero or negative aggregate with a per-row median of +1 is therefore consistent with (a) *support divergence*, where the

two losses place their magnitude on disjoint rows, so the aggregate drifts toward orthogonality without any directional opposition; or with (b) a small number of *high-norm opposed rows* that dominate the weighted sum while the many low-norm rows align. The 0.33% opposed-by-count figure does not bound the opposed *mass*, so counting rows cannot distinguish the two. Separating them requires the per-row gradient *norms*; we measure them directly (Section 4.2, Figure 3) and find (b).

**A four-row illustration.** Take a weight matrix with four rows. Let three low-norm rows be aligned,  $g_i^{\text{ce}} = g_i^{\text{mtp}} = (\epsilon, 0)$  for  $i = 1, 2, 3$  with small  $\epsilon$ , and let one high-norm row be *opposed*,  $g_4^{\text{ce}} = (1, 0)$  and  $g_4^{\text{mtp}} = (-1, 0)$ . Every per-row cosine is defined: three read +1 and one reads -1, so the median and the  $\cos < 0$  fraction (1/4) both say “mostly aligned.” Yet the flattened vectors are dominated by row 4, so the **aggregate cosine is negative** ( $\approx -1$  as  $\epsilon \rightarrow 0$ ). No support divergence is needed: the losses occupy the *same* rows, but one high-norm row is opposed and it sets the sign. This is mechanism (b), and it is what we measure at scale.

### 3.4 Calibration

To confirm the instrument is not trivially reporting alignment, we run a control (A3) that measures CE against an unrelated L1 regularizer on the same head. A well-calibrated instrument should read near-zero alignment for two unrelated losses.

## 4 Experiments and Results

### 4.1 Per-row gradients are predominantly aligned on the rows that carry supervision

**The honest denominator: active rows.** Because CE and the shared MTP loss read the *same* logits from the *same* hidden state, a vocabulary row that is not a target anywhere in the measurement batch has  $g_i^{\text{mtp}} = 0.75 g_i^{\text{ce}}$  exactly (the two objectives differ only in their target-indicator terms, and  $0.75 = 0.5 + 0.25$  is the summed auxiliary weight; the empirically measured norm ratio  $\|g_i^{\text{mtp}}\|/\|g_i^{\text{ce}}\|$  on such rows is 0.750, which is itself a check that all horizons use the same hidden state). Such rows are scalar multiples with cosine identically +1, and they dominate the count. We therefore classify rows by *target presence*, not by a cosine threshold: a row is *active* iff  $|\|g_i^{\text{mtp}}\|/\|g_i^{\text{ce}}\| - 0.75| > 0.01$ . By this construction  $\approx 88\text{--}92\%$  of rows are parallel and only  $\approx 8\text{--}12\%$  **are active** (the exact fraction is batch-relative; see below). Restricted to the active rows, the median per-row cosine is  $\approx 0.53$  (Muon  $0.529 \pm 0.006$ , AdamW  $0.547 \pm 0.013$ ,  $n=3$ ),  $\approx 3\%$  are opposed, and  $\approx 83\%$  exceed  $\cos > 0.3$ . Over the full vocabulary the same run reads a median of +1.00, 0.33% opposed, and 98.4% above  $|\cos| > 0.3$  against a 0.42% calibrated control (Figure 2); these are the construction-inflated numbers, reported only to calibrate against the control on a common denominator. The active-row median of  $\approx 0.53$  is the honest measure of directional alignment, and it is still well above zero, in sharp contrast to the near-zero aggregate. Crucially, the active-row correction does *not* affect the mechanism in Section 4.2, which is mass-weighted: the parallel rows carry only  $\approx 4\%$  of the gradient-magnitude mass, so the norm-profile cosine (0.98) and opposed-norm fraction (0.60/0.69) are identical whether computed over all rows or active rows only. The interference lives among the active rows; restricting to them makes the finding sharper, not weaker.

### 4.2 The aggregate cosine decouples from per-row alignment during training

The aggregate cosine nonetheless reads  $\approx 0$  ( $-0.08$  at step 1000). Tracking both metrics over training (Figure 5) shows this is not a fixed property: at initialization the aggregate and per-row alignment **agree at**  $\approx +0.99$ , and the aggregate then crosses zero early (between steps 100 and 150 under Muon, and between 150 and 200 under AdamW) while the per-row median holds at  $\approx +1$  for the remainder of the 1000-step diagnostic window (late-training persistence is shown separately at 350M, Section 4.5). These trajectories are a single diagnostic run per optimizer; the close agreement between two different optimizers is the replication we have.

We can pin down *how* the aggregate collapses without any additional runs, using an exact identity. The aggregate cosine is a magnitude-weighted average of the per-row cosines,  $\text{agg} = \sum_i w_i \cos_i$  with weights

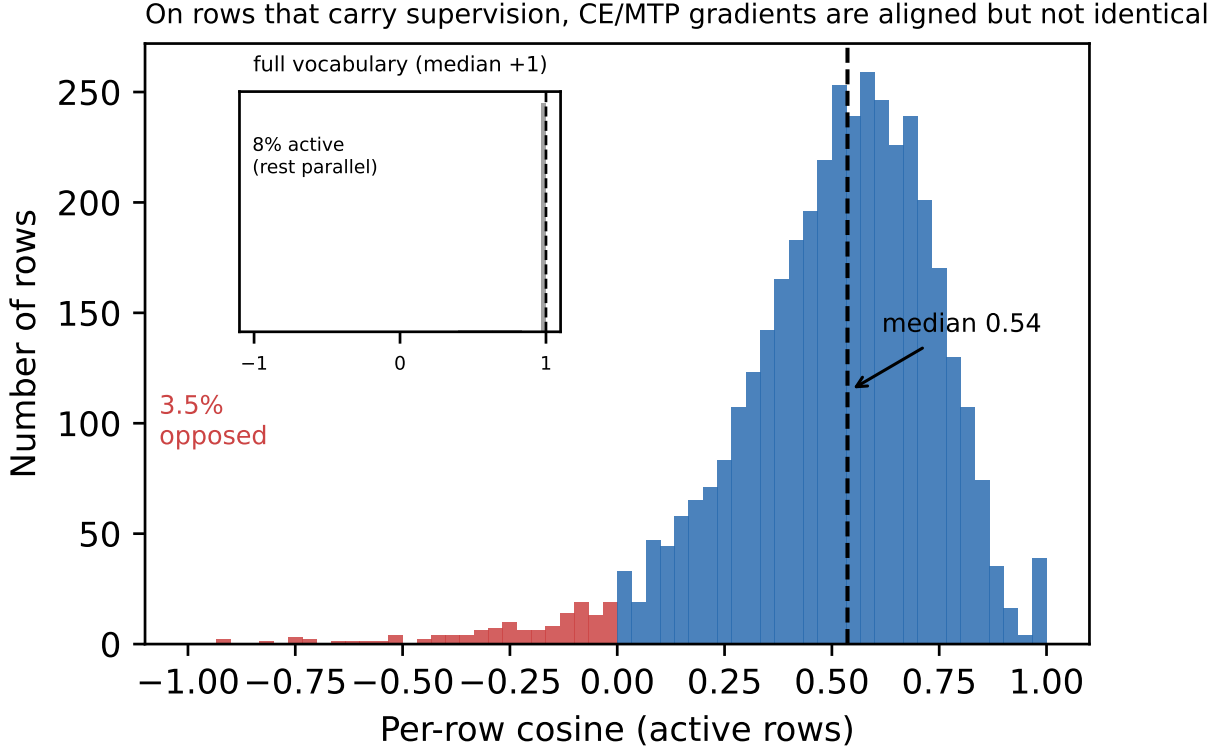


Figure 2: **Per-row CE/MTP gradients on the output projection are aligned on the rows that carry supervision.** Main panel: distribution of *signed* per-row cosines over the active rows (median  $\approx 0.53$ ;  $\approx 3\%$  opposed, in red). Inset: the full-vocabulary distribution, whose median is  $+1.00$  because  $\approx 88\text{--}92\%$  of rows are parallel by construction ( $g^{\text{mtp}} = 0.75 g^{\text{ce}}$  for any row with no target in the batch, since CE and the shared MTP loss read the same logits). Right: a CE-vs-L1 control reads  $0.42\%$  of rows above  $|\cos| > 0.3$  versus  $98.4\%$  full-vocab for CE-vs-MTP; on active rows the comparison is  $\approx 83\%$  versus the control, still a large calibration gap. GPT-2 124M, step 1000, 50,257 vocab rows (47 padding rows masked). The mass-weighted mechanism (Section 4.2) is unaffected by the active-row restriction because parallel rows carry only  $\approx 4\%$  of the gradient mass.

$w_i = \|g_i^{\text{ce}}\| \|g_i^{\text{mtp}}\| / (\|G^{\text{ce}}\| \|G^{\text{mtp}}\|) \geq 0$ . If gradient magnitude were spread evenly across rows, the weights would be equal and the aggregate would equal the *plain mean* of the per-row cosines. That mean is  $+0.94$  (Muon) and  $+0.96$  (AdamW), whereas the actual aggregate is  $-0.08$  and  $-0.28$  (Figure 6). The entire  $\approx 1.0$  gap between “aggregate if norms were flat” and the real aggregate is therefore attributable to the per-row *norm* structure, not to the typical per-row direction: the standard diagnostic is reporting a magnitude-weighting artifact, which is the paper’s central point made quantitative.

The weights are unnormalized, and their sum is itself informative. Writing  $a_i = \|g_i^{\text{ce}}\|$  and  $b_i = \|g_i^{\text{mtp}}\|$ , we have  $\sum_i w_i = \langle a, b \rangle / (\|a\| \|b\|) = \cos(a, b)$ , which is exactly the *norm-profile cosine*  $\rho_{\text{norm}}$  defined below. The aggregate therefore factors exactly as

$$\text{agg} = \sum_i w_i \cos_i = \underbrace{\left( \sum_i w_i \right)}_{\rho_{\text{norm}}} \cdot \underbrace{\frac{\sum_i w_i \cos_i}{\sum_i w_i}}_{\langle \cos \rangle_{\text{mass}}} = \rho_{\text{norm}} \langle \cos \rangle_{\text{mass}}, \quad (1)$$

the product of how much the two losses’ norm profiles overlap and the mass-weighted mean of the per-row cosines. Numerically this is  $-0.08 = 0.98 \times (-0.082)$  (Muon) and  $-0.28 = 0.98 \times (-0.286)$  (AdamW). The two “disambiguating statistics” we introduce next are therefore not competing hypotheses but the two exact factors of the aggregate: support overlap is the total weight  $\rho_{\text{norm}}$ , and opposition is carried by  $\langle \cos \rangle_{\text{mass}}$ .

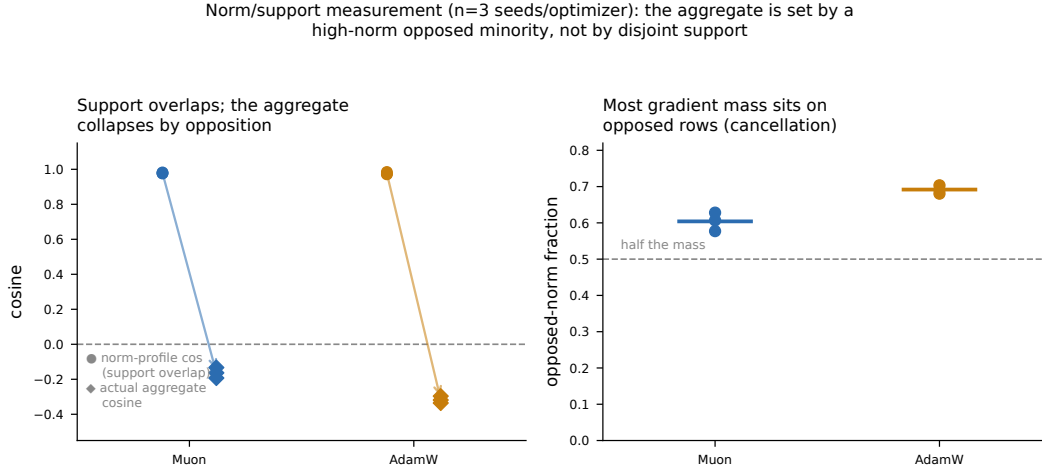


Figure 3: **The aggregate is set by a high-norm opposed minority, not by disjoint support** ( $n=3$  seeds per optimizer, step 1000, 50,257 real vocabulary rows). *Left*: the norm-profile cosine (the cosine of the two per-row gradient-norm vectors) is  $\approx 0.98$  for both optimizers, so CE and MTP load the same rows; support does not diverge. The actual aggregate cosine nonetheless sits negative, and the arrow marks the collapse. *Right*: the opposed-norm fraction, the share of  $\|g_i^{\text{ce}}\| \|g_i^{\text{mtp}}\|$  mass on rows with  $\cos < 0$ , is above one half for both optimizers (0.60 Muon, 0.69 AdamW): the opposed minority carries most of the magnitude mass. AdamW cancels more strongly than Muon; the mechanism is identical, the magnitude optimizer-dependent.

What the identity does *not* settle is which norm structure is responsible. A negative aggregate requires the opposed minority ( $\cos < 0$ ,  $\approx 0.33\%$  of rows) to carry *some* weight, so two mechanisms are a priori compatible with the aligned-median-plus-negative-aggregate pattern: (a) *support divergence*, where CE and MTP load their magnitude onto increasingly disjoint rows, making the products  $w_i$  small precisely where each gradient is large; and (b) a small number of *high-norm opposed rows* dominating the weighted sum. Separating (a) from (b) needs the per-row gradient *norms*, which the earlier runs do not store, so we measure them directly with a drop-in instrument (`measure_norm_support.py`) that logs per-row norms and reports two disambiguating statistics on the same short diagnostic pass: the *norm-profile cosine* (the cosine of the two per-row norm vectors;  $\approx 0$  under support divergence,  $\approx 1$  if the losses load the same rows) and the *opposed-norm fraction* (the share of  $\|g_i^{\text{ce}}\| \|g_i^{\text{mtp}}\|$  mass on rows with  $\cos < 0$ ). Across three seeds per optimizer, the norm-profile cosine is 0.98 (Muon  $0.979 \pm 0.003$ , AdamW  $0.977 \pm 0.007$ ): CE and MTP load the *same* rows, so support does *not* diverge. Yet the opposed-norm fraction is  $0.60 \pm 0.03$  (Muon) and  $0.69 \pm 0.01$  (AdamW): the majority of the magnitude mass sits on the opposed minority. Mechanism (b) holds, and it holds under both optimizers, with AdamW cancelling more strongly than Muon (Figure 3); the difference in opposed-norm fraction (0.69 vs 0.60) is significant across seeds (Welch  $t=5.4$ ,  $p \approx 0.015$ ), so the magnitude is genuinely optimizer-dependent while the mechanism is shared. The near-zero aggregate is a norm-weighted cancellation by a high-norm opposed minority, not disjoint support.

**Which rows carry the cancellation.** The opposed mass is not spread thinly: on the Muon seed-42 pass, 140 opposed rows (0.28% of the vocabulary) carry 60.7% of the total  $\|g_i^{\text{ce}}\| \|g_i^{\text{mtp}}\|$  mass, and just 47 rows account for two-thirds of *all* gradient mass (Figure 4, left). Decoding those rows shows they are the most frequent function words and punctuation: the four highest-mass opposed rows are `.`, the newline, `the`, and `,`, which alone carry  $\approx 41\%$  of the mass, followed by `of`, `to`, `and`, and `a` (Figure 4, right). This is the expected locus: these tokens are ubiquitous next-token *and* future targets, and their next-token and future-token continuation distributions genuinely differ, so the CE and MTP gradients on their output rows point in different directions while both being large. The cancellation is concentrated on a handful of high-frequency rows, not diffuse across the tail.



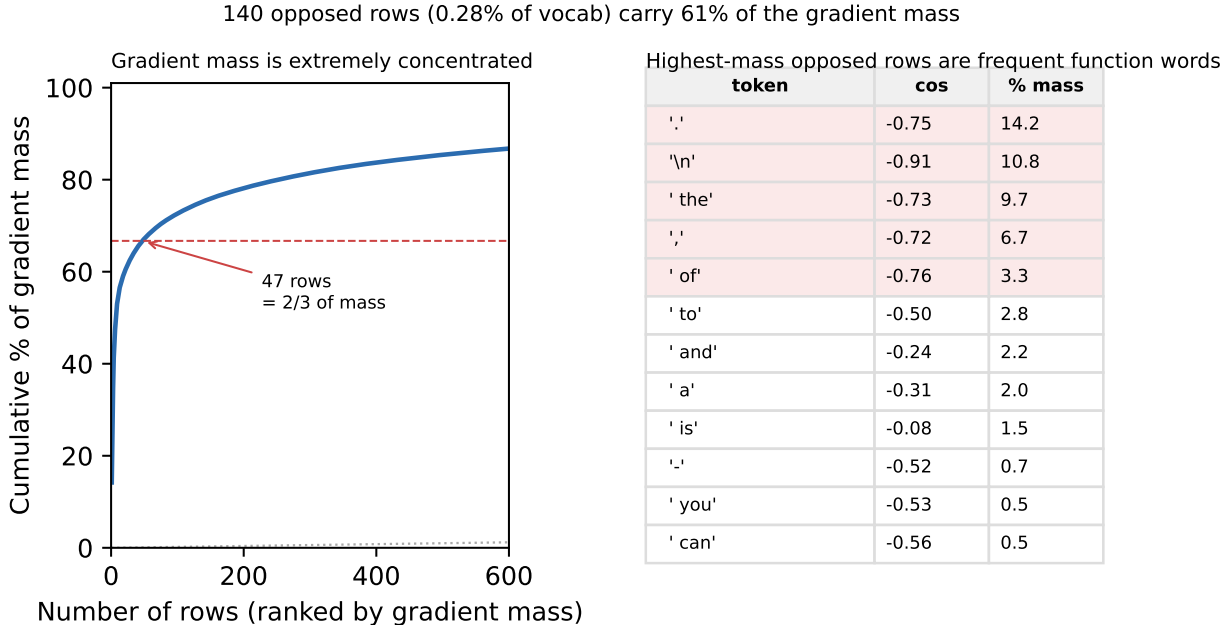


Figure 4: **The cancellation is carried by a few high-frequency rows.** *Left:* Lorenz curve of the per-row gradient mass ( $\|g_i^{\text{ce}}\| \|g_i^{\text{mtp}}\|$ , ranked descending); 47 rows account for two-thirds of all mass (dashed line), against the line of equality (dotted). *Right:* the highest-mass opposed rows, decoded, are the most frequent function words and punctuation; the top four carry  $\approx 41\%$  of the total mass. Muon, seed 42, step 1000.

### 4.3 The pattern is general, and survives untying

High per-row alignment is **not unique to the output head**. Across all 74 weight matrices (Figure 7), attention and MLP matrices also show high per-row alignment (many above 0.8), frequently with strongly *negative* aggregate cosine (down to  $-0.78$ ). The output head is distinctive only in being the point of **maximum** per-row alignment (0.984) sitting nearest aggregate-zero, and, more importantly, in being the matrix that *directly produces the logits*. The head matters not because the metric singles it out, but because interference there is paid directly in next-token loss. Alignment also **persists when the head is untied** (A4: 99.3% of rows above threshold, versus 98.7% tied at the matched step), so it is a property of the output-projection gradient ( $h^\top \partial \mathcal{L} / \partial \text{logits}$ ), not an artifact of weight tying. These two percentages are *full-vocabulary* and therefore construction-floored ( $\geq 75\%$  of rows are parallel by construction); the untied run did not commit per-row norms, so we cannot restate them on the active-row denominator, and we report them only as a point check that untying does not *reduce* alignment. The evidentiary weight of this check is modest in any case: the soft-label identity (Section 4.4) predicts untied-persistence *a priori*, since that identity never references tying. The per-layer generality (Figure 7) and the untied-head comparison are each *single-run* measurements at a fixed step; we report them as point observations, not seed-replicated estimates (only the 124M CE-vs-MTP loss gap is replicated across seeds).

We are deliberate about what this generality does and does not license. The same qualitative signature (high per-row alignment with a non-positive aggregate) appears across the network, and at many matrices the aggregate is *strongly* negative (to  $-0.78$ ), not merely near zero. This means the per-row/aggregate gap is a fairly generic property of these structured gradient matrices, so a near-zero aggregate on its own is *not* a specific signature of two-loss interference. What is specific to the output head is that it is the one matrix whose per-row gradients are the CE and MTP gradients directly, whose norm-weighted opposition we measure end to end (Figure 3), and whose interference is paid immediately in next-token loss. We do not adjudicate the per-row norm structure of the strongly-negative aggregates elsewhere in the network; a per-row account of those is beyond what our measurements settle.

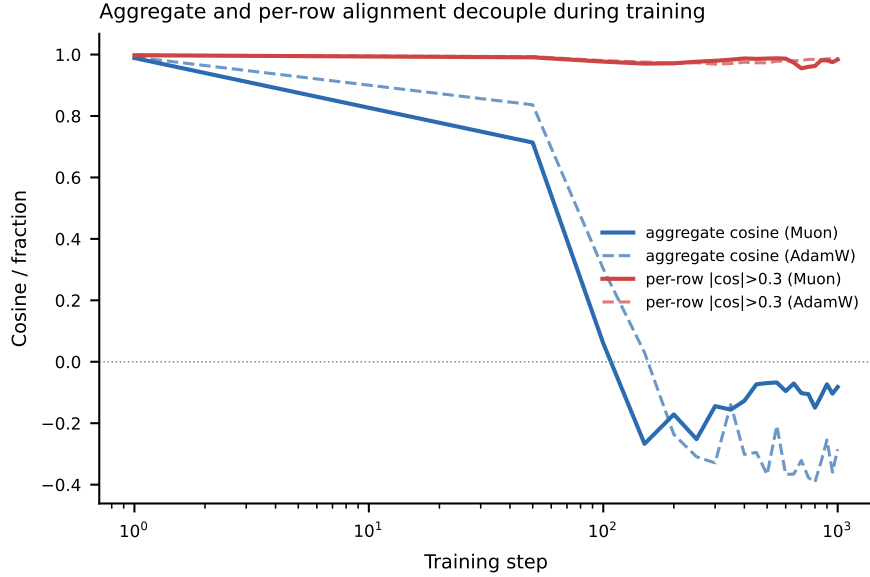


Figure 5: **The aggregate and per-row alignment start together and decouple.** Both metrics begin at  $\approx +0.99$ ; within  $\sim 200$  steps the aggregate (matrix-level) cosine falls below zero (crossing between steps 100–150 for Muon and 150–200 for AdamW) while per-row alignment stays high. The per-row curve is the full-vocabulary  $|\cos| > 0.3$  fraction ( $\approx 0.98$ ); on active rows the level is lower ( $\approx 0.83$ , Section 4.1) but the decoupling from the aggregate is identical: the aggregate goes negative under either denominator. One diagnostic run per optimizer (Muon solid, AdamW dashed); the agreement between two optimizers is the replication.

#### 4.4 Adding shared MTP degrades next-token loss; no auxiliary recovers it

Adding the shared MTP objective degrades next-token validation loss from  $3.985 \pm 0.014$  (CE-only,  $n=5$ ) to  $4.377 \pm 0.021$  ( $n=3$ ): a  $+0.39$  nat ( $\approx 10\%$ ) increase. On the three seeds the two conditions share (42, 123, 456), a paired test gives a mean gap of **0.382** nats (per-seed 0.397/0.380/0.370),  $t=50.0$ ,  $p \approx 4.0 \times 10^{-4}$  (df=2); treating the conditions as independent (Welch, CE-only minus shared-MTP, using all  $n=5$  and  $n=3$  seeds) gives  $t = -28.8$  (negative because CE-only is the lower loss),  $p \approx 8 \times 10^{-5}$  with a 95% CI on the difference of [0.35, 0.43] nats. The gap is large relative to the seed-to-seed spread (within-condition sd  $\approx 0.01$ –0.02), which is what makes it detectable at these small  $n$ ; we report the effect as the 0.39-nat loss increase itself rather than a standardized effect size, since the latter is inflated by the tight reproducibility of a controlled testbed. With  $n=3$  per condition and this spread, the minimum difference we have  $\approx 80\%$  power to detect is  $\Delta \approx 0.05$  nats; the headline gap is  $\sim 8\times$  that floor, so it is detected with room to spare, whereas differences below  $\approx 0.05$  nats (relevant to the tuned auxiliary below) are at or under our detection limit. We report uncorrected  $p$ -values for the  $k=5$  variant-vs-baseline comparisons; under a Holm correction at  $\alpha=0.05$  the shared-MTP, stop-gradient, standard-contrastive, and NextLat results remain significant, while only the tuned-auxiliary comparison ( $p \approx 0.19$ ) does not survive, consistent with its effect sitting below the detection floor. Detaching the auxiliary from the head via stop-gradient makes it *worse* (4.493,  $\Delta+0.508$ ,  $p < 10^{-4}$ ), not better, which points to a norm-weighting/capacity account rather than directional conflict (we note this is a data point, not a proof: stop-gradient also removes the auxiliary’s ability to shape the shared representation, which is a competing reason it could hurt). The  $n=5$  CE-only and  $n=3$  shared-MTP conditions share three seeds (42, 123, 456); the two extra CE-only seeds (789, 1337) were run for a tighter baseline and no MTP seeds were dropped. Capacity-separated auxiliaries reduce but do **not** recover the baseline. The *standard* layer-split contrastive auxiliary ( $G_{\text{ncc}}$ : an InfoNCE-style contrastive loss applied to an intermediate layer, giving the auxiliary its own representation capacity instead of sharing the output head) reaches 4.024 ( $\Delta+0.039$ ,  $p \approx 0.010$ ), statistically distinguishable from A, i.e. a genuine non-recovery.

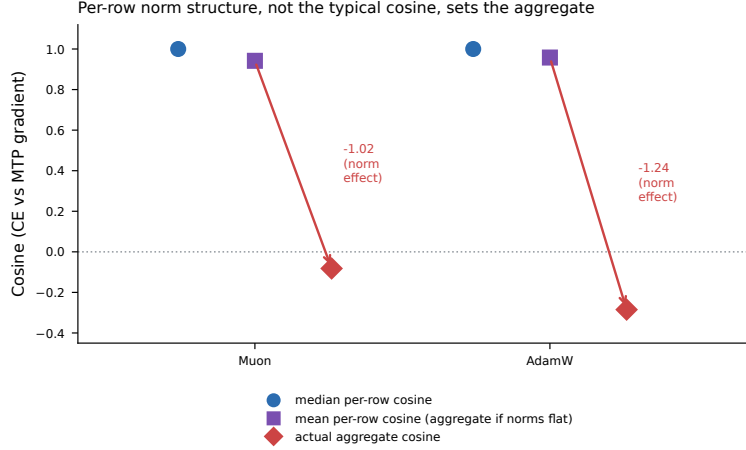


Figure 6: **Per-row norm structure, not the typical cosine, sets the aggregate.** For each optimizer: the median per-row cosine (+1.00), the plain *mean* per-row cosine (which equals the aggregate the standard diagnostic would report if gradient magnitude were spread evenly across rows), and the *actual* aggregate cosine. The mean is strongly positive (+0.94/+0.96) while the actual aggregate is negative (−0.08/−0.28); the gap is due entirely to per-row norm weighting. Single diagnostic run per optimizer, step 1000, 50,257 real vocabulary rows.

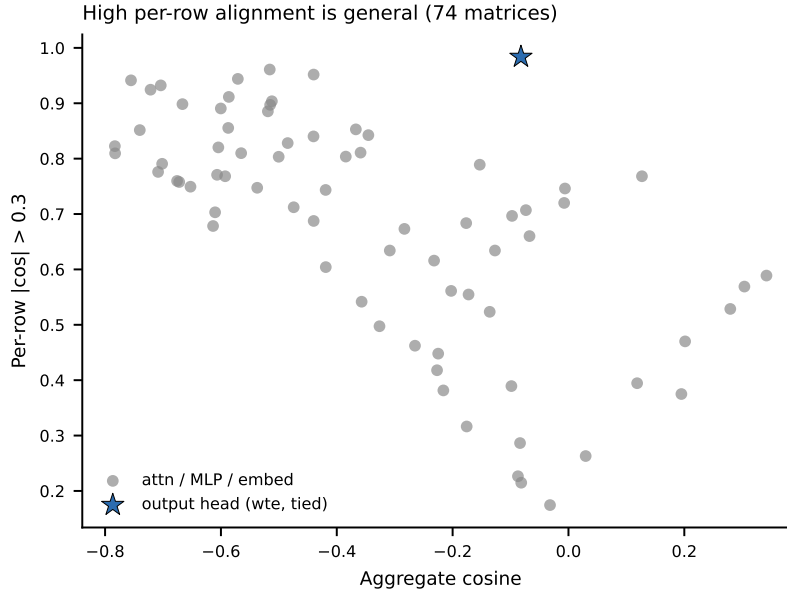


Figure 7: **High per-row alignment is general across the network.** Per-row alignment vs aggregate cosine for all 74 weight matrices. The output head (star) is the maximum and sits nearest aggregate-zero, but attention/MLP matrices show the same pattern, many with strongly negative aggregate cosine.

A hyperparameter-tuned variant reaches 4.003 ( $\Delta+0.018$ ) but is *not* statistically separable from A at  $n=3$  ( $p \approx 0.19$ ): for this variant we can claim neither recovery nor non-recovery, and we rest the capacity conclusion on the standard variant. A latent-prediction baseline (NextLat: the auxiliary predicts the *hidden state* at future offsets rather than their tokens, so it never touches the output head) lands at 4.037 ( $\Delta+0.051$ , Welch  $p \approx 0.018$ ) (Figure 8).

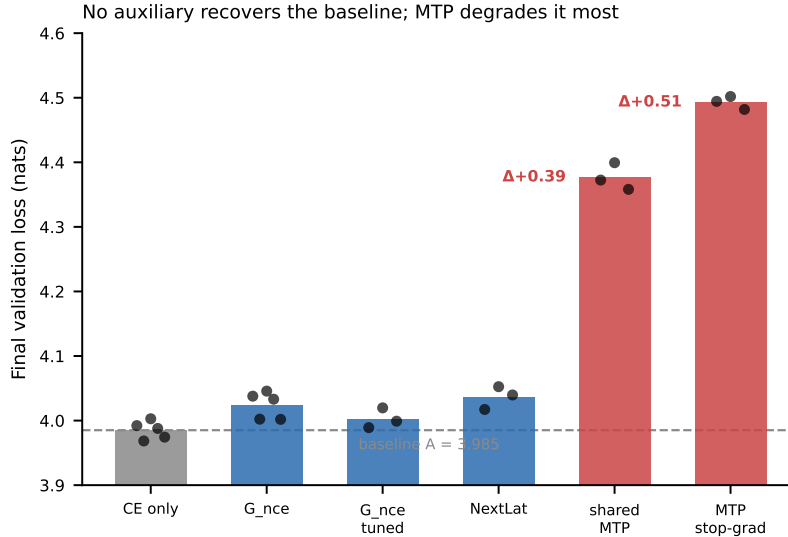


Figure 8: **No auxiliary recovers the baseline; MTP degrades it most.** Final next-token validation loss by variant (points = seeds). Shared MTP degrades CE by  $\Delta+0.39$ ; stop-grad by  $\Delta+0.51$ ; no capacity-separated auxiliary recovers the CE-only baseline within noise.

Conflict-targeted gradient surgery also does not recover the baseline, and this is a stronger statement than “there was nothing to remove.” The opposed rows are negligible by count (0.3%) but carry the majority of the gradient mass (60–69%), and a projection acts on the mass-weighted vector, so surgery had a great deal to project away; per-row Gram–Schmidt on the head targets exactly those opposed rows and still recovers nothing. The measured opposition is mass-dominant and causally inert. Because those runs are single-seed with a numerical-drift caveat, we report them in full in Appendix A rather than resting any claim on them.

#### 4.5 The apparent rare-token alignment is a construction artifact; the pattern holds at 350M

Measured over the full vocabulary, per-row alignment appears *higher* for rarer tokens: the rarest quintile averages 0.98 mean  $|\cos|$  versus 0.86 for the most-common quintile (Figure 9, left). This effect is an *artifact of the construction floor*. Rare tokens are almost never targets in a measurement batch, so they are overwhelmingly parallel-by-construction ( $\cos = +1$ ): the active fraction falls from  $\approx 25\%$  in the common quintile to  $\approx 1.5\%$  in the rarest. Restricted to active rows, the apparent gradient disappears: active-row mean  $|\cos|$  is essentially flat across frequency quintiles (0.51–0.54 for Muon, 0.52–0.60 for AdamW; Figure 9, right), with only a weak residual (Spearman  $\rho \approx 0.12$  between token id and active-row  $|\cos|$ ). We therefore *retract* the rare-token-alignment reading: the full-vocabulary ordering reflects how many rows are parallel by construction, not a frequency-dependent alignment. The aligned-per-row-with-negative-aggregate picture holds qualitatively at **350M** (single seed): CE-only 3.752, shared-MTP 4.129 ( $\Delta+0.378$ ).

The 350M runs are the paper’s only *late-training* per-row measurement, logged out to step 87,000. The instrument computes the same quantity in both runs (the per-row cosine between the CE gradient and the MTP gradient on the output projection) regardless of which objective the model was trained with (the MTP gradient is computed at measurement time for both). Measured this way, the shared-MTP-trained model holds  $|\cos| > 0.3$  at 98.9% of rows at step 87,000, and the CE-only-trained model reads 94.5% at the same step. These are *full-vocabulary* figures and thus construction-floored: with  $\geq 75\%$  of rows parallel by construction, the informative range is the remaining  $\approx 25$  percentage points, within which the MTP-trained model sits near the top and the CE-only model  $\approx 4$  points lower (coarse, but not vacuous), stated here at its true resolution (the 350M runs committed only summary statistics, so an active-row restatement is not recoverable from the saved data). That the CE-only model *also* shows high CE/MTP alignment this late indicates the alignment is an intrinsic property of the output-projection gradient geometry, not something the

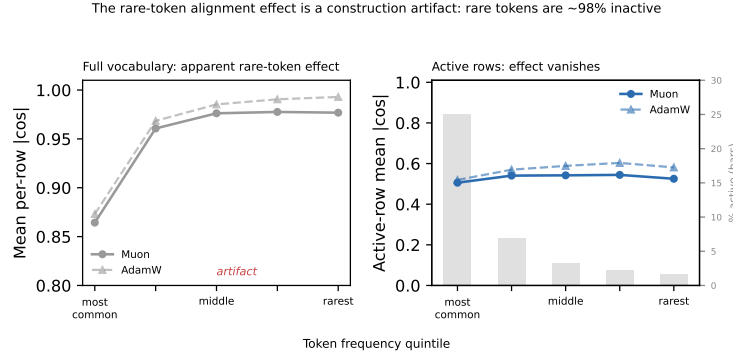


Figure 9: **The rare-token alignment effect is a construction artifact.** Left: over the full vocabulary, mean per-row  $|\cos|$  rises from common to rare tokens (both optimizers); this is the apparent effect. Right: restricted to active rows (those carrying a target), the effect vanishes and mean  $|\cos|$  is flat across frequency quintiles; the grey bars show the active fraction collapsing from  $\approx 25\%$  (common) to  $\approx 1.5\%$  (rare), which is what drives the full-vocabulary trend. Single run per optimizer, step 1000, GPT-2 124M.

MTP objective induces. Both are single seeds, so we treat the per-row persistence as illustrative (consistent with alignment not decaying over a long horizon), not established. The 350M *loss* gap ( $\Delta +0.378$  above) is a next-token-loss number, unaffected by the active-row denominator, and it is this scale-stability of the gap ( $+0.392$  at 124M,  $+0.378$  at 350M) that Section 4.4 builds on.

## 5 Discussion

**What the measurements establish.** In this regime (a single shared tied output projection,  $\leq 350M$ , undertrained) the standard cosine conflict diagnostic misreads the CE/MTP interaction. On the rows that carry a target the per-row gradients are aligned (active-row median cosine  $\approx 0.53$ ,  $\approx 3\%$  opposed; the full-vocabulary median is  $+1$  only because  $\approx 88\text{--}92\%$  of rows are scalar multiples by construction), the aggregate is  $\approx 0$  or negative, and the two decouple during training. Instrumenting the per-row norms shows the aggregate is set by a high-norm opposed minority ( $\approx 0.3\%$  of all rows, carrying 60–69% of the mass), not by disjoint support: the losses load the same rows (norm-profile cosine 0.98) but oppose on the few rows that dominate the weighted sum. Crucially, the next-token degradation is large and unambiguous, yet neutralizing that directional opposition does not remove it: the surgery baselines in Appendix A recover nothing, so the opposed minority the cosine detects is not the fixable lever. The lever that helps is capacity separation, though in our runs it reduces rather than eliminates the gap.

**Two failure modes, one instrument.** The aggregate cosine misleads in two opposite directions depending on the optimizer, and the paper’s central claim is that both are failures of the *same* diagnostic. Under Muon the aggregate is  $\approx -0.08$ , close enough to zero that a conflict diagnostic reads “no interference,” a *false negative*: the next-token loss degrades by 0.39 nats regardless. Under AdamW the aggregate is  $\approx -0.28$ , negative enough to trip a conflict flag, but this is a *misleading positive*: gradient surgery that projects out exactly this opposition recovers nothing (Appendix A), because the opposition is mass-dominant yet causally inert. A practitioner reading only the aggregate would conclude “safe to add MTP” under Muon and “apply surgery” under AdamW, and both conclusions are wrong. The instrument does not merely lose power; it points the wrong way, and which way depends on an optimizer choice unrelated to whether interference is present. This is why the per-row and per-norm decomposition, not a scalar threshold on the aggregate, is what the regime requires.

**What they do not establish.** We do not show that MTP *as practiced* (with separate per-horizon heads) degrades next-token modeling; our formulation is a single tied head, and separate-head designs give the auxiliary its own capacity precisely so that the losses cannot compete on shared rows. We do not establish

causation between the norm-weighted opposition and the loss increase; we show they co-occur and that interventions behave as a norm-weighting/capacity account predicts (in particular, surgery that removes the opposition does not remove the degradation). And the effect lives in an undertrained, small-batch,  $\leq 350\text{M}$  regime where MTP is already documented to underperform (Zuhri et al., 2025).

**Future work.** Three experiments would broaden the scope from “the diagnostic misreads this regime” toward a causal, general statement: (1) a **separate-head MTP** variant, to localize whether the degradation is intrinsic to head-sharing; (2)  $n=3$  **at 350M** plus a 124M CE-only per-row baseline, to turn the single-seed scale observation into a measured trend; and (3) **matched-seed, single-pass gradient surgery** at  $n=3$ , to remove the numerical-drift caveat. A further question, *which* norm structure produces the collapse, is settled directly here: instrumenting the per-row norms (`measure_norm_support.py`,  $n=3$  seeds per optimizer) shows the two losses load the same rows (norm-profile cosine 0.98) while a high-norm opposed minority carries the majority of the mass (opposed-norm fraction 0.60/0.69), so the mechanism is norm-weighted cancellation, not support divergence (Section 4.2, Figure 3).

## 6 Conclusion

Cosine similarity between task gradients is the default instrument for detecting interference in multi-task and auxiliary-loss training. We showed that at a language model’s shared, tied output projection this instrument is misleading: the per-row CE and MTP gradients are aligned in direction on the rows that carry a target (active-row median cosine  $\approx 0.53$ ; the full-vocabulary median is  $+1$  because most rows are scalar multiples by construction), the aggregate cosine is driven negative by a high-norm opposed minority ( $\approx 0.3\%$  of rows carrying most of the mass) rather than by a broad directional conflict, and the two metrics decouple during training. Adding the shared auxiliary degrades next-token loss substantially, and neutralizing the measured opposition does not repair it, so the degradation is not the directional conflict a cosine diagnostic would report. The aggregate is set by per-row gradient norms, not by the per-row directions the cosine averages, a distinction that explains why conflict-targeted surgery does not help here and why capacity separation does. We deliver the result on a fully reproducible testbed with a committed statistics script, and we are explicit about the shared-head, small-scale scope and the experiments that would extend it.

## Reproducibility Statement

All results are computed from committed result JSONs by `analysis/stats.py`; the seven data figures are regenerated from the same JSONs by `figures/make_figures.py` (the conceptual schematic, Figure 1, by `figures/make_schematic.py`). Code, data-loading, pinned dependencies (`requirements.txt`), and a README with exact run commands are released. The optimizer (Muon) is pinned to a specific commit. Scope and caveats are stated in Section 1 and Section 5.

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## A Gradient-surgery baselines (single-seed; reported for completeness)

We ran conflict-targeted gradient-surgery methods on the shared-MTP variant to test directly whether removing directional conflict recovers the baseline. **We do not rest any claim in the main text on these runs**, for three reasons stated up front: (i) each is a **single seed** (seed 42); (ii) the PCGrad baseline operates on **all parameters**, whereas per-row Gram-Schmidt and Scatter operate on the **output head**

**only**, so they are not strictly comparable; and (iii) surgery uses **two backward passes** under bf16 (versus the fused single pass of the main runs), introducing a  $\sim 0.02\text{--}0.04$  nat numerical drift comparable to the smaller surgery effects. With those caveats, the pattern is uniform: no surgery method recovers the CE-only baseline ( $A = 3.985$ ), and none improves meaningfully on the matched shared-MTP seed ( $B@42 = 4.390$ ).

| Method                     | Acts on    | Final val loss | $\Delta$ vs B@42 | $\Delta$ vs A |
|----------------------------|------------|----------------|------------------|---------------|
| CE only (A, ref)           | —          | 3.985          | —                | 0.000         |
| shared MTP (B@42)          | —          | 4.390          | 0.000            | +0.405        |
| Per-row Gram–Schmidt       | head only  | 4.418          | +0.027           | +0.432        |
| Scatter                    | head only  | 4.458          | +0.067           | +0.472        |
| PCGrad                     | all params | 4.544          | +0.154           | +0.559        |
| Pure-Muon (all 2D in Muon) | all params | 4.556          | +0.166           | +0.571        |

Table 1: Gradient-surgery baselines on shared MTP (seed 42). Every method leaves the loss at or above the shared-MTP baseline; the per-row Gram–Schmidt effect (+0.027) sits inside the two-pass drift envelope, so we treat it as “no recovery,” not as a measured improvement.

This is what a norm-weighting/capacity account predicts, and it is stronger than “nothing to project away.” The opposed rows carry the majority of the gradient-magnitude mass (60–69%; Section 4.2), and a projection acts on that mass-weighted vector, so there was substantial opposition to remove; per-row Gram–Schmidt on the head targets exactly those rows and still leaves the loss at the shared-MTP baseline. The measured opposition is mass-dominant yet causally inert. A matched-seed, single-pass replication at  $n=3$  (identified in Section 5) would remove the drift caveat and is the right way to upgrade this from a consistent-with observation to evidence. The AdamW negatives from the same phase ( $A = 3.743/3.754$ ,  $B = 4.149/4.153$ ,  $G_{\text{ncc}} = 3.778/3.781$  across seeds 42/123) reproduce the ordering under a second optimizer and are included in `results/phase_c/`.